



End Term (Even) Semester Regular/Back examination May 2026

Roll no. 2318681

Name of the Program and semester: B.Tech Core/AI-ML/AI-DS/CS VI SEM
Name of the Course : Full Stack Web Development
Course Code: TCS 693
Time: 3 hours

Maximum Marks: 100

Note:

- (i) All the questions are compulsory.
- (ii) Answer any two sub questions from a, b and c in each question.
- (iii) Total marks for each question is 20 (twenty).
- (iv) Each sub-question carries 10 marks.

- (2X10=20 Marks)
- Q1.
- a) Explain in brief what is meant by the term Semantic Elements in HTML. Why are they important, and how do they contribute to the structure and accessibility of a webpage? CO1
 - b) Describe how CSS frameworks (like Bootstrap) help in web development. Explain the importance of responsive design and how CSS supports it. CO1
 - c) Describe the CSS box model in detail. Also, explain pseudo-classes and pseudo-elements in CSS, demonstrating the usage of :hover, :nth-child(), :before, and :after. CO1

- (2X10=20 Marks)
- Q2.
- a) Define DTD (Document Type definition). What is a PCDATA and CDATA section in XML? With the help of XML, create a database of employees and validate it with the help of DTD. CO2
 - b) What is the difference between client-side and server-side form validation? Write a JavaScript program that validates a form with fields for Name, Email, Phone Number, and Password, displaying error messages for invalid inputs using regular expression. CO1, CO2
 - c) What are JavaScript events? Explain different types of events such as click, mouseover, keypress, and focus. Write a program that Change Color Randomly on Mouse Hover over a rectangular region. CO2

- (2X10=20 Marks)
- Q3.
- a) Explain difference between GET and POST method in form handling. Also explain different methods of date object. Display the current date and check whether today is a weekend or weekday. CO2
 - b) Explain how jQuery simplifies DOM manipulation and event handling. Write a jQuery script to create a simple slideshow that automatically cycles through images stored in an array. CO2
 - c) Explain the concept of backend data loading and also, write a program to retrieve JSON data using AJAX. CO3

- (2X10=20 Marks)
- Q4.
- a) Design an HTML form to accept the following student details: Student ID, Student Name, Course, Year, and Email. Write a PHP program to connect to a MySQL database and insert the submitted



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data into an existing table named students. Display an appropriate success message if the record is inserted successfully; otherwise, display an error message. CO4

b) Explain the concept of cookies and session in web development. Write a PHP script to Set a cookie named "user_pref" with an expiry time of 1 day. Also retrieve and display the stored preference. CO4

c) What is file handling in PHP? Explain different file modes (read, write, append). Describe how data can be read from an existing file and displayed dynamically on a webpage CO4, CO5
(2X10=20 Marks)

Q5.

a) Define the MEAN Stack. Explain each of its components—MongoDB, Express.js, Angular, and Node.js in detail. Describe how CRUD operations are performed in a MEAN Stack application. CO5

b) Define React.js and explain its core principles. Describe component-based architecture and Virtual DOM in detail. CO5

c) What is a Content Management System (CMS)? Describe the key features of a CMS, such as content creation, editing, publishing, and user management. CO5